



ARTIFICIAL VISION

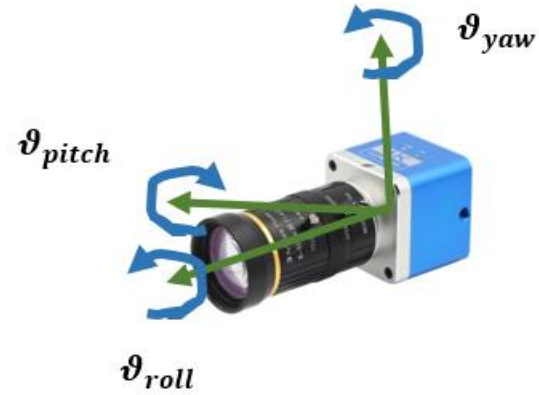
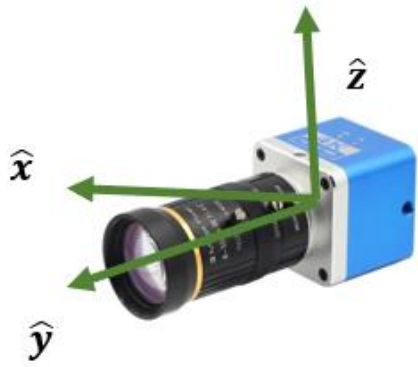
2024/2025
EXERCISE ON 3D/2D MAPPING

EXERCISE

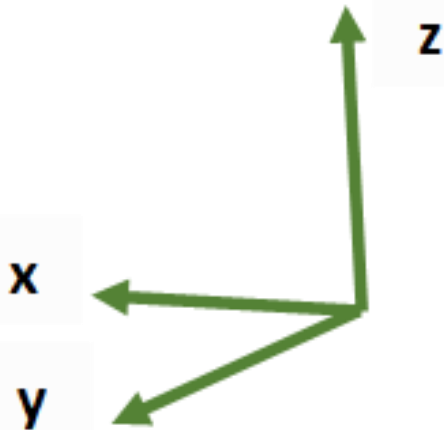
Given the information about the camera parameters and position, translate the given coordinates of 4 points in image coordinates.



EXERCISE



Camera reference system



World reference system

CASE 1 – TEST20241212.PNG

```
image_file = "test20241212.png"  
f = 0.003  
U = 1280  
V = 720  
thyaw = 0*pi/180  
throll = 0*pi/180  
thpitch = -32*pi/180  
xc = 0  
yc = 0  
zc = 7.20  
  
x_real = [-2.5, 0.5, 0.5, 4.6]  
y_real = [13.41, 8.00, 13.00, 10.91]  
  
s_w = 0.00498  
s_h = 0.00374
```



CASE 2 – TEST.PNG

```
image_file = "test.png"
f = 0.00325
U = 1920
V = 1080
thyaw = 12*pi/180
throll = 10.5*pi/180
thpitch = -36*pi/180
xc = 0
yc = 0
zc = 6.92

x_real = [-3.5, 3.6, -2.8, 2]
y_real = [13.11, 10.61, 4.11, 5.81]

s_w = 0.00498
s_h = 0.00374
```

